

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login Status · Short Circuiting (React)</title>
  <!-- React & ReactDOM -->
  <script crossorigin src="https://unpkg.com/react@18.2.0/umd/react.development.js"></script>
  <script crossorigin src="https://unpkg.com/react-dom@18.2.0/umd/react-dom.development.js"></script>
  <!-- Babel for JSX -->
  <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
</head>
<style>
  * {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
  }

  body {
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    padding: 20px;
  }

  .container {
    background: white;
    border-radius: 30px;
    padding: 3rem;
    box-shadow: 0 30px 60px rgba(0,0,0,0.3);
    max-width: 500px;
    width: 100%;
    text-align: center;
  }

  h1 {
    color: #2d3748;
    font-size: 2.2rem;
    margin-bottom: 0.5rem;
  }

  .subtitle {
    color: #718096;
    margin-bottom: 2.5rem;
    font-size: 1.1rem;
    border-bottom: 2px solid #e2e8f0;
    padding-bottom: 1rem;
  }

  .status-card {
    background: #f7fafc;
    border-radius: 20px;
    padding: 2rem;
    margin: 2rem 0;
    border: 2px solid #e2e8f0;
  }

  .status-message {
    font-size: 1.8rem;
    font-weight: 600;
    padding: 1.5rem;
    border-radius: 15px;
    border: 1px solid #e2e8f0;
    transition: all 0.3s ease;
  }

  .logged-in {
    background: #c6f6d5;
    color: #22543d;
    border: 2px solid #48bb78;
  }

  .logged-out {
    background: #fed7d7;
    color: #742a2a;
    border: 2px solid #f56565;
  }

  .button-group {
    display: flex;
    gap: 1rem;
    justify-content: center;
    margin-top: 2rem;
  }

  .btn {
    padding: 1rem 2rem;
    font-size: 1.1rem;
    font-weight: 600;
    border: none;
    border-radius: 50px;
    cursor: pointer;
    transition: all 0.2s;
    flex: 1;
    text-transform: uppercase;
    letter-spacing: 0.5px;
  }

  .btn-login {
    background: #4299e1;
    color: white;
    box-shadow: 0 4px 6px rgba(66, 153, 225, 0.4);
  }

  .btn-login:hover {
    background: #3182ce;
    transform: translateY(-2px);
  }

  .btn-logout {
    background: #f56565;
    color: white;
    box-shadow: 0 4px 6px rgba(245, 101, 101, 0.4);
  }

  .btn-logout:hover {
    background: #e53e3e;
    transform: translateY(-2px);
  }

  .code-snippet {
    background: #2d3748;
    color: #a0aec0;
    padding: 1rem;
    border-radius: 15px;
    font-family: 'Courier New', monospace;
    font-size: 0.9rem;
    margin-top: 2rem;
    text-align: left;
  }

  .code-snippet.highlight {
    color: #68d391;
    font-weight: bold;
  }

  .output-badge {
    background: #edf2f7;
    padding: 0.5rem 1.5rem;
    border-radius: 40px;
    display: inline-block;
    margin-bottom: 1rem;
    font-size: 0.9rem;
    color: #4a5568;
    border: 1px dashed #cbd5e0;
  }

  .short-circuit-note {
    background: #ffffff;
    border-left: 5px solid #4299e1;
    padding: 1rem;
    border-radius: 10px;
    margin: 1.5rem 0;
    text-align: left;
  }

  .short-circuit-note code {
    background: #e2e8f0;
    padding: 0.2rem 0.5rem;
    border-radius: 5px;
    color: #2d3748;
  }
</style>
```

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font-weight: 600;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1> Login Status</h1>
    <div class="subtitle">With Short-Circuit Evaluation (&&)</div>

    <!-- React mount point -->
    <div id="root"></div>

    <div class="code-snippet">
      <div style="color: #fff; margin-bottom: 0.5rem;">< Short-circuit logic:</div>
      <span style="color: #fbbf6d;">{isLoggedIn}</span> <span style="color: #fff;">&&</span> <span style="color: #68d391;">"You are logged in"</span><span style="color: #fff;">"</span>
      <div style="color: #00a08a; margin-top: 0.5rem; font-size: 0.8rem;">
        // Message renders ONLY when isLoggedIn = true
      </div>
    </div>
  </div>

  <script type="text/babel">
    // LoginStatus Component using short-circuiting
    const LoginStatus = () => {
      // State variable for login status
      const [isLoggedIn, setIsLoggedIn] = React.useState(false);

      return (
        <div>
          { /* Output badge showing current state */ }
          <div className="output-badge">
            Current output state: isLoggedIn = {isLoggedIn ? 'true' : 'false'}
          </div>

          { /* Status Card with Short-circuit rendering */ }
          <div className="status-card">
            <div className={ `status-message ${isLoggedIn ? 'logged-in' : 'logged-out'} ` }>
              { /* SHORT-CIRCUITING IN ACTION */ }
              {isLoggedIn && "You are logged in"}
              { /* When isLoggedIn is false, this renders nothing (empty) */ }
              { !isLoggedIn && "You are logged out" }
            </div>

            { /* Small note showing what's rendering */ }
            <div className="short-circuit-note">
              <strong> Short-circuit used:</strong><br />
              <code>{isLoggedIn && "You are logged in"}</code>
              {isLoggedIn ? (
                <span style={{color: '#48bb78', display: 'block', marginTop: '0.5rem'}}>
                  ✔ Since isLoggedIn is true, it shows: "You are logged in"
                </span>
              ) : (
                <span style={{color: '#f56565', display: 'block', marginTop: '0.5rem'}}>
                  ❌ Since isLoggedIn is false, it shows nothing (short-circuited)
                </span>
              ) }
            </div>

            { /* Buttons to toggle login status */ }
            <div className="button-group">
              <button
                className="btn btn-login"
                onClick={() => setIsLoggedIn(true)}
                disabled={!isLoggedIn}
                style={{opacity: isLoggedIn ? 0.5 : 1}}
              >
                Login
              </button>
              <button
                className="btn btn-logout"
                onClick={() => setIsLoggedIn(false)}
                disabled={isLoggedIn}
                style={{opacity: !isLoggedIn ? 0.5 : 1}}
              >
                Logout
              </button>
            </div>

            { /* Additional explanation for PDF/output screenshot */ }
            <div style={{marginTop: '2rem', fontSize: '0.9rem', color: '#718096', borderTop: '1px solid #e2e8f0', padding: '1rem'}}>
              <strong> Expected Output Screenshot Shows:</strong><br/>
              • When Logged Out: Empty message area (short-circuit prevents rendering)<br/>
              • When Logged In: Green box with "You are logged in" message<br/>
              • Boolean state toggles via Login/Logout buttons
            </div>
          </div>
        </div>
      );
    };

    // Render the component
    const root = ReactDOM.createRoot(document.getElementById('root'));
    root.render(<LoginStatus />);
  </script>

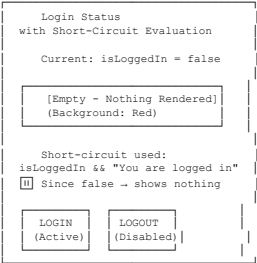
  <!-- Floating indicator for output visualization -->
  <div style="position: fixed; bottom: 20px; right: 20px; background: #2d3748; color: white; padding: 0.5rem 1.5rem; border-radius: 50px; font-size: 0.8rem; box-shadow: 0 5px 15px rgba(0,0,0,0.3); border: 2px solid #667eea;">
    $ Short-circuit: {isLoggedIn && "Message"} → Shows only when true
  </div>
</body>
</html>
```

Expected Output (Visual Description)

When you run this code, you'll see two different states:

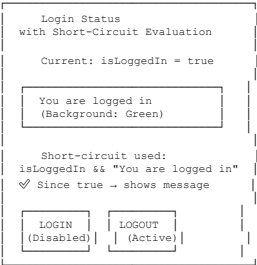
State 1: Logged Out (Initial State)

text



State 2: Logged In (After clicking Login)

text



Key Points Shown in Output:

1. **Short-circuit in action:** `{isLoggedIn && "You are logged in"}`

- When `false` → renders nothing (empty)
- When `true` → renders "You are logged in"

2. **Visual feedback:**

- Red background when logged out (shows empty state)
- Green background with message when logged in

3. **Interactive buttons:**

- Login button enables when logged out
- Logout button enables when logged in
- Buttons are disabled in opposite state